



Statistical Distributions · Mark Scheme

Statistics Chapter 6: Statistical Distributions. M = method mark A = accuracy mark (depends on M) B = independent mark ft = follow through

Section A – Discrete random variables

- 1 (a) $k = 1 - 0.7 = 0.3$ B1; (b) $1 - 0.2 = 0.8$ B1; (c) $0.3 + 0.3 = 0.6$ B1.
2 $k(1 + 2 + 3 + 4) = 1$ M1; $k = 0.1$ A1; $P(X > 2) = 0.3 + 0.4 = 0.7$ A1.
3 $0.1 + a = 0.45 \Rightarrow a = 0.35$ M1 A1; $b = 1 - 0.65 = 0.35$ A1.
4 (a) $P(X = 1) = \frac{1}{16}$, $P(X = 2) = \frac{3}{16}$ M1 A1; $P(X = 3) = \frac{5}{16}$, $P(X = 4) = \frac{7}{16}$ A1. (b) $\frac{12}{16} = \frac{3}{4}$ B1.
5(a) $k(4 + 3 + 2 + 1) = 1$ M1; $10k = 1 \Rightarrow k = 0.1$ A1.
5(b) Probabilities 0.4, 0.3, 0.2, 0.1 B1.
5(c) 0.7 B1.
5(d) Pairs (1,4), (4,1), (2,3), (3,2) M1; $2(0.4 \times 0.1) + 2(0.3 \times 0.2)$ M1; $= 0.2$ A1.

Section B – The binomial distribution

- 6 $\binom{10}{3}(0.3)^3(0.7)^7$ M1; $= 0.2668$ A1.
7 (a) 0.1268 B1; (b) 0.2173 B1; (c) $1 - P(X \leq 7) = 0.2131$ B1.
8 (a) $50 \times 0.2 = 10$ B1; (b) $X \sim B(50, 0.2)$: $P(X = 10) = 0.1398$ B1; $P(X \leq 8) = 0.3073$ B1.
9 (a) 0.2454 B1; (b) $1 - P(X \leq 7)$ M1; $= 0.3990$ A1; (c) $P(X \leq 8) - P(X \leq 3) = 0.7180$ B1.
10(a) Seeds germinate independently of each other B1; the probability of germinating is the same (0.85) for every seed B1.
10(b) $X \sim B(20, 0.85)$ M1; $P(X = 18) = 0.2293$ A1.
10(c) $P(X \leq 15)$ M1; $= 0.1702$ A1.
10(d) $1 - P(X \leq 16)$ M1; $= 0.6477$ A1.
10(e) $Y \sim B(6, 0.6477)$, $P(Y \geq 5) = 1 - P(Y \leq 4)$ M1; $= 0.3148$ A1.

Section C – The normal distribution

- 11 $P(Z < 1)$ M1; $= 0.8413$ A1.
12 (a) $P(Z > 1.333)$ M1; $= 0.0912$ A1; (b) within one sd: 0.6827 B1.
13 (a) $P(-1 < Z < 2)$ M1; $= 0.8186$ A1; (b) 0.0228 B1.
14 (a) $\mu \pm 2\sigma$ M1; 26 to 34 minutes A1. (b) 0, since X is continuous B1.
15(a) $P(X > 170) = P(Z > 1.667)$ M1; $= 0.0478$ A1.
15(b) $P(-0.833 < Z < 1.25)$ M1; $= 0.692$ A1.
15(c) $1 - (1 - 0.0478)^{10}$ M1; $= 0.387$ A1.
15(d) $\frac{w - 150}{12} = -1.2816$ M1; $w = 134.6$ g A1.

Section D – The inverse normal

- 16 $z = 1.2816$ B2 (B1 for 1.28).
17 $\frac{a - 60}{8} = 1.6449$ M1 B1; $a = 73.2$ A1.



- 18 $\frac{55 - \mu}{5} = 1.2816$ M1 B1; $\mu = 48.6$ A1.
- 19 $P(Z < z) = 0.945 \Rightarrow z = 1.598$ B1; $\frac{40 - 30}{\sigma} = 1.598$ M1 A1; $\sigma = 6.26$ A1.
- 20(a) $\frac{35 - \mu}{\sigma} = -1.2816$ M1 B1; $\frac{70 - \mu}{\sigma} = 1.6449$ B1; solving: $\sigma = 12.0$ M1; $\mu = 50.3$ A1.
- 20(b) $P(40 < X < 55)$ with $X \sim N(50.3, 12.0^2)$ M1; standardising correctly M1; $= 0.458$ A1.

Section E – Choosing and evaluating a model

- 21 (a) Binomial B1; (b) normal B1.
- 22 Cards are not replaced B1, so the probability of an ace changes each time: the trials are not independent B1.
- 23 Mean $\approx$ median, so roughly symmetrical B1; nearly all data within $\mu \pm 3\sigma$ (23.9 to 26.3) B1; length is a continuous variable B1.
- 24 $P(X < 0) = P(Z < -1.25) = 0.106$ M1 A1; waiting times cannot be negative, so over 10% of the model is impossible: not suitable B1.
- 25(a) $P(Z < -0.5625)$ M1; $= 0.287$ A1; rainfall cannot be negative, so the model gives a large probability to impossible values B1.
- 25(b) Positively skewed: most days have little or no rain, a few days have a lot B1.
- 25(c) $W \sim B(7, 0.35)$ M1; $P(W \geq 4) = 1 - P(W \leq 3)$ M1; $= 0.200$ A1; assumption: days are wet independently of each other (with constant probability) B1.

Roar-level challenge

- 26 $1 - 0.85^n > 0.99$ M1; $0.85^n < 0.01 \Rightarrow n > \frac{\ln 0.01}{\ln 0.85} = 28.3$ M1 A1; $n = 29$ (check: $1 - 0.85^{28} = 0.9894$, $1 - 0.85^{29} = 0.9910$) A1.